

lab11

Sam Medina

```
library(bio3d)
```

Warning: package 'bio3d' was built under R version 4.5.3

```
library(bio3dview)

results_dir <- "novelProtein_7a997"
pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names=TRUE)
basename(pdb_files)
```

```
[1] "novelProtein_7a997_unrelaxed_rank_001_alphafold2_ptm_model_3_seed_000.pdb"
[2] "novelProtein_7a997_unrelaxed_rank_002_alphafold2_ptm_model_2_seed_000.pdb"
[3] "novelProtein_7a997_unrelaxed_rank_003_alphafold2_ptm_model_5_seed_000.pdb"
[4] "novelProtein_7a997_unrelaxed_rank_004_alphafold2_ptm_model_1_seed_000.pdb"
[5] "novelProtein_7a997_unrelaxed_rank_005_alphafold2_ptm_model_4_seed_000.pdb"
```

```
pdb<- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

Reading PDB files:

```
novelProtein_7a997/novelProtein_7a997_unrelaxed_rank_001_alphafold2_ptm_model_3_seed_000.pdb
novelProtein_7a997/novelProtein_7a997_unrelaxed_rank_002_alphafold2_ptm_model_2_seed_000.pdb
novelProtein_7a997/novelProtein_7a997_unrelaxed_rank_003_alphafold2_ptm_model_5_seed_000.pdb
novelProtein_7a997/novelProtein_7a997_unrelaxed_rank_004_alphafold2_ptm_model_1_seed_000.pdb
novelProtein_7a997/novelProtein_7a997_unrelaxed_rank_005_alphafold2_ptm_model_4_seed_000.pdb
.....
```

Extracting sequences


```

151 . . . . 200

201 . . . . 250
[Truncated_Name:1]novelProte IPRQPRCPLTEAGSTQATWPSGARTATSSRTDRKTSSFRVGRYPPWRWR
[Truncated_Name:2]novelProte IPRQPRCPLTEAGSTQATWPSGARTATSSRTDRKTSSFRVGRYPPWRWR
[Truncated_Name:3]novelProte IPRQPRCPLTEAGSTQATWPSGARTATSSRTDRKTSSFRVGRYPPWRWR
[Truncated_Name:4]novelProte IPRQPRCPLTEAGSTQATWPSGARTATSSRTDRKTSSFRVGRYPPWRWR
[Truncated_Name:5]novelProte IPRQPRCPLTEAGSTQATWPSGARTATSSRTDRKTSSFRVGRYPPWRWR
*****
201 . . . . 250

251 . . . . 300
[Truncated_Name:1]novelProte ASSTSTPRSLRRPSWHATTPSGVSL LAPSWSLGRAPRRSSTGGAGRTW
[Truncated_Name:2]novelProte ASSTSTPRSLRRPSWHATTPSGVSL LAPSWSLGRAPRRSSTGGAGRTW
[Truncated_Name:3]novelProte ASSTSTPRSLRRPSWHATTPSGVSL LAPSWSLGRAPRRSSTGGAGRTW
[Truncated_Name:4]novelProte ASSTSTPRSLRRPSWHATTPSGVSL LAPSWSLGRAPRRSSTGGAGRTW
[Truncated_Name:5]novelProte ASSTSTPRSLRRPSWHATTPSGVSL LAPSWSLGRAPRRSSTGGAGRTW
*****
251 . . . . 300

301 . . . . 350
[Truncated_Name:1]novelProte SRTWCPAPSLWTPCPRQAPARSRSTCSDLAPRSSPSRLGQAPSFDPHPC
[Truncated_Name:2]novelProte SRTWCPAPSLWTPCPRQAPARSRSTCSDLAPRSSPSRLGQAPSFDPHPC
[Truncated_Name:3]novelProte SRTWCPAPSLWTPCPRQAPARSRSTCSDLAPRSSPSRLGQAPSFDPHPC
[Truncated_Name:4]novelProte SRTWCPAPSLWTPCPRQAPARSRSTCSDLAPRSSPSRLGQAPSFDPHPC
[Truncated_Name:5]novelProte SRTWCPAPSLWTPCPRQAPARSRSTCSDLAPRSSPSRLGQAPSFDPHPC
*****
301 . . . . 350

351 . . . . 400
[Truncated_Name:1]novelProte GLPPRRQVHGRGLLPRQSRHEGLPEESRGNQDVLRLVPLRRLGRLGRGR
[Truncated_Name:2]novelProte GLPPRRQVHGRGLLPRQSRHEGLPEESRGNQDVLRLVPLRRLGRLGRGR
[Truncated_Name:3]novelProte GLPPRRQVHGRGLLPRQSRHEGLPEESRGNQDVLRLVPLRRLGRLGRGR
[Truncated_Name:4]novelProte GLPPRRQVHGRGLLPRQSRHEGLPEESRGNQDVLRLVPLRRLGRLGRGR
[Truncated_Name:5]novelProte GLPPRRQVHGRGLLPRQSRHEGLPEESRGNQDVLRLVPLRRLGRLGRGR
*****
351 . . . . 400

401 . . . . 450
[Truncated_Name:1]novelProte LHPAEGPIERHHHFGWGEHILPGGGEHPLQAPQDRGGRRGTRRHQVGVSL
[Truncated_Name:2]novelProte LHPAEGPIERHHHFGWGEHILPGGGEHPLQAPQDRGGRRGTRRHQVGVSL
[Truncated_Name:3]novelProte LHPAEGPIERHHHFGWGEHILPGGGEHPLQAPQDRGGRRGTRRHQVGVSL
[Truncated_Name:4]novelProte LHPAEGPIERHHHFGWGEHILPGGGEHPLQAPQDRGGRRGTRRHQVGVSL

```

```

[Truncated_Name:5]novelProte  LHPAEGPIERHHHFGWGEHILPGGGEHPLQAPQDRGGRRGTRRHQVGVS
*****
401 . . . . 450

451 . 469
[Truncated_Name:1]novelProte  RLLGAEAWAERHRGGALQV
[Truncated_Name:2]novelProte  RLLGAEAWAERHRGGALQV
[Truncated_Name:3]novelProte  RLLGAEAWAERHRGGALQV
[Truncated_Name:4]novelProte  RLLGAEAWAERHRGGALQV
[Truncated_Name:5]novelProte  RLLGAEAWAERHRGGALQV
*****
451 . 469

```

Call:

```
pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")
```

Class:

```
pdb, fasta
```

Alignment dimensions:

```
5 sequence rows; 469 position columns (469 non-gap, 0 gap)
```

```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

```
rd<-rmsd(pdb,fit=T)
```

Warning in rmsd(pdb, fit = T): No indices provided, using the 469 non NA positions

```
range(rd)
```

```
[1] 0.000 41.879
```

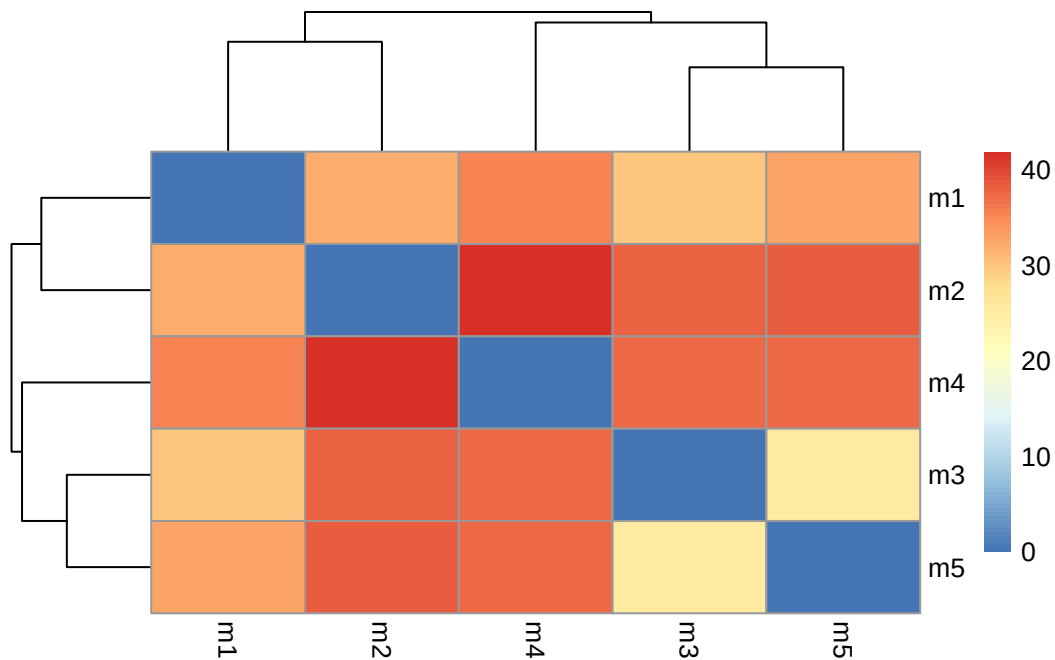
```
library(pheatmap)
```

Warning: package 'pheatmap' was built under R version 4.5.3

```

colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)

```



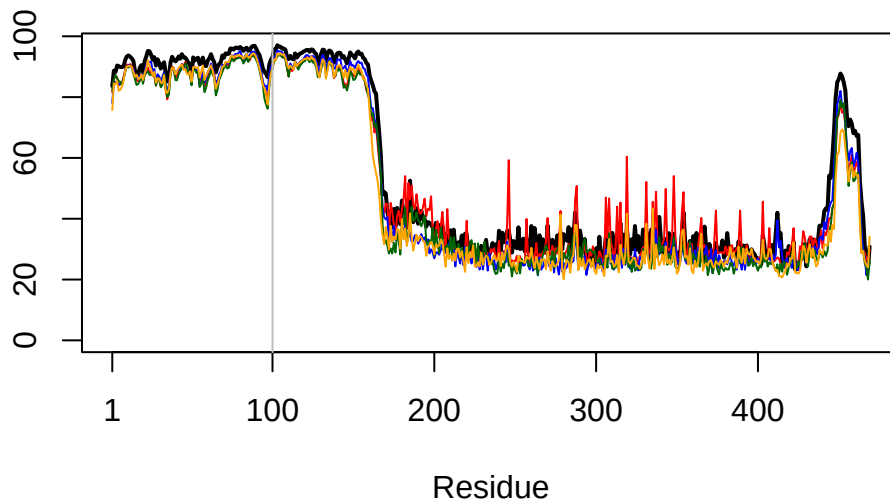
```
pdb<-read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
plotb3(pdb$b[1,], typ="l", lwd=2, sse=pdb)
```

Warning in plotb3(pdb\$b[1,], typ = "l", lwd = 2, sse = pdb): Length of input 'sse' does not equal the length of input 'x'; Ignoring 'sse'

```
points(pdb$b[2,], typ="l", col="red")
points(pdb$b[3,], typ="l", col="blue")
points(pdb$b[4,], typ="l", col="darkgreen")
points(pdb$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```



```
core<-core.find(pdbbs)
```

```
core size 468 of 469 vol = 2852673
core size 467 of 469 vol = 2862624
core size 466 of 469 vol = 2841275
core size 465 of 469 vol = 2810853
core size 464 of 469 vol = 2777028
core size 463 of 469 vol = 2742301
core size 462 of 469 vol = 2706845
core size 461 of 469 vol = 2673077
core size 460 of 469 vol = 2638677
core size 459 of 469 vol = 2607566
core size 458 of 469 vol = 2576668
core size 457 of 469 vol = 2546980
core size 456 of 469 vol = 2516547
core size 455 of 469 vol = 2488565
core size 454 of 469 vol = 2463248
core size 453 of 469 vol = 2434735
core size 452 of 469 vol = 2410086
core size 451 of 469 vol = 2386879
core size 450 of 469 vol = 2363729
core size 449 of 469 vol = 2339843
```

core size 448 of 469 vol = 2316075
core size 447 of 469 vol = 2293112
core size 446 of 469 vol = 2266558
core size 445 of 469 vol = 2241629
core size 444 of 469 vol = 2216043
core size 443 of 469 vol = 2189473
core size 442 of 469 vol = 2163754
core size 441 of 469 vol = 2134438
core size 440 of 469 vol = 2107606
core size 439 of 469 vol = 2080296
core size 438 of 469 vol = 2052147
core size 437 of 469 vol = 2020491
core size 436 of 469 vol = 1992238
core size 435 of 469 vol = 1967760
core size 434 of 469 vol = 1938612
core size 433 of 469 vol = 1908960
core size 432 of 469 vol = 1880089
core size 431 of 469 vol = 1849900
core size 430 of 469 vol = 1819257
core size 429 of 469 vol = 1796202
core size 428 of 469 vol = 1769062
core size 427 of 469 vol = 1743323
core size 426 of 469 vol = 1717734
core size 425 of 469 vol = 1692161
core size 424 of 469 vol = 1666735
core size 423 of 469 vol = 1641261
core size 422 of 469 vol = 1620824
core size 421 of 469 vol = 1597172
core size 420 of 469 vol = 1577196
core size 419 of 469 vol = 1553023
core size 418 of 469 vol = 1529430
core size 417 of 469 vol = 1506147
core size 416 of 469 vol = 1484901
core size 415 of 469 vol = 1467672
core size 414 of 469 vol = 1450852
core size 413 of 469 vol = 1433860
core size 412 of 469 vol = 1418047
core size 411 of 469 vol = 1395963
core size 410 of 469 vol = 1373905
core size 409 of 469 vol = 1356920
core size 408 of 469 vol = 1339267
core size 407 of 469 vol = 1323468
core size 406 of 469 vol = 1301562

core size 405 of 469 vol = 1280849
core size 404 of 469 vol = 1265981
core size 403 of 469 vol = 1250409
core size 402 of 469 vol = 1235252
core size 401 of 469 vol = 1214371
core size 400 of 469 vol = 1197858
core size 399 of 469 vol = 1183266
core size 398 of 469 vol = 1165017
core size 397 of 469 vol = 1146703
core size 396 of 469 vol = 1131045
core size 395 of 469 vol = 1117500
core size 394 of 469 vol = 1102972
core size 393 of 469 vol = 1088556
core size 392 of 469 vol = 1075008
core size 391 of 469 vol = 1060544
core size 390 of 469 vol = 1046264
core size 389 of 469 vol = 1030091
core size 388 of 469 vol = 1014750
core size 387 of 469 vol = 999285.1
core size 386 of 469 vol = 988073.5
core size 385 of 469 vol = 973417.1
core size 384 of 469 vol = 954524.9
core size 383 of 469 vol = 938510
core size 382 of 469 vol = 923274.6
core size 381 of 469 vol = 911451.4
core size 380 of 469 vol = 897998.6
core size 379 of 469 vol = 883052.4
core size 378 of 469 vol = 869225.4
core size 377 of 469 vol = 855330.2
core size 376 of 469 vol = 842484.3
core size 375 of 469 vol = 831440.5
core size 374 of 469 vol = 820513.8
core size 373 of 469 vol = 809400
core size 372 of 469 vol = 796405.6
core size 371 of 469 vol = 786851.3
core size 370 of 469 vol = 768872.3
core size 369 of 469 vol = 754718.4
core size 368 of 469 vol = 740905.7
core size 367 of 469 vol = 727390.8
core size 366 of 469 vol = 713381.6
core size 365 of 469 vol = 701177.4
core size 364 of 469 vol = 688516.4
core size 363 of 469 vol = 675640.9

core size 362 of 469 vol = 662546.4
core size 361 of 469 vol = 650283.5
core size 360 of 469 vol = 639494.6
core size 359 of 469 vol = 627020.8
core size 358 of 469 vol = 616165.8
core size 357 of 469 vol = 605811.3
core size 356 of 469 vol = 593653.6
core size 355 of 469 vol = 581874.4
core size 354 of 469 vol = 570364
core size 353 of 469 vol = 560026.4
core size 352 of 469 vol = 549803.6
core size 351 of 469 vol = 539028.9
core size 350 of 469 vol = 528822.5
core size 349 of 469 vol = 518892.5
core size 348 of 469 vol = 507430.5
core size 347 of 469 vol = 494974.4
core size 346 of 469 vol = 485747
core size 345 of 469 vol = 473759.9
core size 344 of 469 vol = 463590.6
core size 343 of 469 vol = 453352.8
core size 342 of 469 vol = 440489.5
core size 341 of 469 vol = 429970.5
core size 340 of 469 vol = 419877
core size 339 of 469 vol = 406995.7
core size 338 of 469 vol = 394493.1
core size 337 of 469 vol = 380399.1
core size 336 of 469 vol = 368611.9
core size 335 of 469 vol = 359416
core size 334 of 469 vol = 350998.1
core size 333 of 469 vol = 342256.9
core size 332 of 469 vol = 334268.8
core size 331 of 469 vol = 323970.9
core size 330 of 469 vol = 316070.4
core size 329 of 469 vol = 308370.1
core size 328 of 469 vol = 301357.9
core size 327 of 469 vol = 290397.8
core size 326 of 469 vol = 283525.4
core size 325 of 469 vol = 274732.8
core size 324 of 469 vol = 269822
core size 323 of 469 vol = 264257
core size 322 of 469 vol = 258403.2
core size 321 of 469 vol = 252413.5
core size 320 of 469 vol = 242279.1

core size 319 of 469 vol = 236117.1
core size 318 of 469 vol = 230217
core size 317 of 469 vol = 225700.1
core size 316 of 469 vol = 222110.7
core size 315 of 469 vol = 217361.2
core size 314 of 469 vol = 212182.4
core size 313 of 469 vol = 205111.3
core size 312 of 469 vol = 200015.6
core size 311 of 469 vol = 195001.9
core size 310 of 469 vol = 185600.8
core size 309 of 469 vol = 179100.9
core size 308 of 469 vol = 174369.7
core size 307 of 469 vol = 173812.4
core size 306 of 469 vol = 173812.9
core size 305 of 469 vol = 173684.2
core size 304 of 469 vol = 173713.8
core size 303 of 469 vol = 172210.7
core size 302 of 469 vol = 171498.9
core size 301 of 469 vol = 169297.1
core size 300 of 469 vol = 168884.6
core size 299 of 469 vol = 168815.9
core size 298 of 469 vol = 168035.1
core size 297 of 469 vol = 163507.5
core size 296 of 469 vol = 158826.8
core size 295 of 469 vol = 157258.3
core size 294 of 469 vol = 155746.1
core size 293 of 469 vol = 153640.6
core size 292 of 469 vol = 151190
core size 291 of 469 vol = 142304.4
core size 290 of 469 vol = 137610.5
core size 289 of 469 vol = 126751.3
core size 288 of 469 vol = 115868.4
core size 287 of 469 vol = 108067.9
core size 286 of 469 vol = 103823.4
core size 285 of 469 vol = 100761
core size 284 of 469 vol = 97308.32
core size 283 of 469 vol = 93789.14
core size 282 of 469 vol = 90408.74
core size 281 of 469 vol = 87096.05
core size 280 of 469 vol = 82256.96
core size 279 of 469 vol = 77678.68
core size 278 of 469 vol = 74340.44
core size 277 of 469 vol = 70527.01

core size 276 of 469 vol = 68337.24
core size 275 of 469 vol = 65759.24
core size 274 of 469 vol = 63237.64
core size 273 of 469 vol = 61298.23
core size 272 of 469 vol = 60022.82
core size 271 of 469 vol = 58756.44
core size 270 of 469 vol = 56819.64
core size 269 of 469 vol = 55247.7
core size 268 of 469 vol = 53862.49
core size 267 of 469 vol = 52051.47
core size 266 of 469 vol = 50400.22
core size 265 of 469 vol = 48674.02
core size 264 of 469 vol = 46903.63
core size 263 of 469 vol = 45048.78
core size 262 of 469 vol = 43152.65
core size 261 of 469 vol = 40985.09
core size 260 of 469 vol = 38446.73
core size 259 of 469 vol = 36411.78
core size 258 of 469 vol = 34385.46
core size 257 of 469 vol = 32402.02
core size 256 of 469 vol = 30593.75
core size 255 of 469 vol = 28830.58
core size 254 of 469 vol = 26983.6
core size 253 of 469 vol = 25173.86
core size 252 of 469 vol = 23565.51
core size 251 of 469 vol = 22002.27
core size 250 of 469 vol = 20431.67
core size 249 of 469 vol = 19032.26
core size 248 of 469 vol = 17897.71
core size 247 of 469 vol = 16873.56
core size 246 of 469 vol = 15755.43
core size 245 of 469 vol = 14638.98
core size 244 of 469 vol = 13628.86
core size 243 of 469 vol = 12754.5
core size 242 of 469 vol = 11965.45
core size 241 of 469 vol = 11053.88
core size 240 of 469 vol = 10249.97
core size 239 of 469 vol = 9474.27
core size 238 of 469 vol = 8667.942
core size 237 of 469 vol = 7860.116
core size 236 of 469 vol = 7093.181
core size 235 of 469 vol = 6360.539
core size 234 of 469 vol = 5735.685

core size 233 of 469 vol = 5283.996
core size 232 of 469 vol = 4792.203
core size 231 of 469 vol = 4432.774
core size 230 of 469 vol = 3996.22
core size 229 of 469 vol = 3569.902
core size 228 of 469 vol = 3316.722
core size 227 of 469 vol = 3119.185
core size 226 of 469 vol = 2886.932
core size 225 of 469 vol = 2701.524
core size 224 of 469 vol = 2464.925
core size 223 of 469 vol = 2243.23
core size 222 of 469 vol = 2061.336
core size 221 of 469 vol = 1864.484
core size 220 of 469 vol = 1666.488
core size 219 of 469 vol = 1521.222
core size 218 of 469 vol = 1371.857
core size 217 of 469 vol = 1265.586
core size 216 of 469 vol = 1151.633
core size 215 of 469 vol = 1018.816
core size 214 of 469 vol = 866.659
core size 213 of 469 vol = 739.137
core size 212 of 469 vol = 609.391
core size 211 of 469 vol = 511.189
core size 210 of 469 vol = 440.212
core size 209 of 469 vol = 387.225
core size 208 of 469 vol = 332.888
core size 207 of 469 vol = 291.868
core size 206 of 469 vol = 245.748
core size 205 of 469 vol = 207.815
core size 204 of 469 vol = 176.6
core size 203 of 469 vol = 151.289
core size 202 of 469 vol = 128.639
core size 201 of 469 vol = 102.752
core size 200 of 469 vol = 85.479
core size 199 of 469 vol = 73.036
core size 198 of 469 vol = 61.459
core size 197 of 469 vol = 50.031
core size 196 of 469 vol = 39.406
core size 195 of 469 vol = 28.376
core size 194 of 469 vol = 23.244
core size 193 of 469 vol = 17.945
core size 192 of 469 vol = 14.043
core size 191 of 469 vol = 12.309

core size 190 of 469	vol = 11.58
core size 189 of 469	vol = 10.947
core size 188 of 469	vol = 10.069
core size 187 of 469	vol = 9.659
core size 186 of 469	vol = 8.975
core size 185 of 469	vol = 8.479
core size 184 of 469	vol = 8.145
core size 183 of 469	vol = 7.779
core size 182 of 469	vol = 7.519
core size 181 of 469	vol = 7.305
core size 180 of 469	vol = 6.981
core size 179 of 469	vol = 6.713
core size 178 of 469	vol = 6.536
core size 177 of 469	vol = 6.295
core size 176 of 469	vol = 6.094
core size 175 of 469	vol = 5.88
core size 174 of 469	vol = 5.689
core size 173 of 469	vol = 5.51
core size 172 of 469	vol = 5.275
core size 171 of 469	vol = 5.102
core size 170 of 469	vol = 4.901
core size 169 of 469	vol = 4.71
core size 168 of 469	vol = 4.556
core size 167 of 469	vol = 4.372
core size 166 of 469	vol = 4.172
core size 165 of 469	vol = 4.032
core size 164 of 469	vol = 3.888
core size 163 of 469	vol = 3.749
core size 162 of 469	vol = 3.562
core size 161 of 469	vol = 3.369
core size 160 of 469	vol = 3.205
core size 159 of 469	vol = 3.05
core size 158 of 469	vol = 2.931
core size 157 of 469	vol = 2.839
core size 156 of 469	vol = 2.763
core size 155 of 469	vol = 2.67
core size 154 of 469	vol = 2.6
core size 153 of 469	vol = 2.531
core size 152 of 469	vol = 2.466
core size 151 of 469	vol = 2.398
core size 150 of 469	vol = 2.344
core size 149 of 469	vol = 2.296
core size 148 of 469	vol = 2.269

```

core size 147 of 469 vol = 2.215
core size 146 of 469 vol = 2.166
core size 145 of 469 vol = 2.15
core size 144 of 469 vol = 2.11
core size 143 of 469 vol = 2.064
core size 142 of 469 vol = 2.003
core size 141 of 469 vol = 1.958
core size 140 of 469 vol = 1.891
core size 139 of 469 vol = 1.797
core size 138 of 469 vol = 1.73
core size 137 of 469 vol = 1.65
core size 136 of 469 vol = 1.568
core size 135 of 469 vol = 1.522
core size 134 of 469 vol = 1.491
core size 133 of 469 vol = 1.454
core size 132 of 469 vol = 1.398
core size 131 of 469 vol = 1.335
core size 130 of 469 vol = 1.278
core size 129 of 469 vol = 1.194
core size 128 of 469 vol = 1.127
core size 127 of 469 vol = 1.081
core size 126 of 469 vol = 1.026
core size 125 of 469 vol = 0.968
core size 124 of 469 vol = 0.906
core size 123 of 469 vol = 0.836
core size 122 of 469 vol = 0.755
core size 121 of 469 vol = 0.698
core size 120 of 469 vol = 0.637
core size 119 of 469 vol = 0.576
core size 118 of 469 vol = 0.52
core size 117 of 469 vol = 0.456
FINISHED: Min vol ( 0.5 ) reached

```

```
core.inds <- print(core, vol=0.5)
```

```

# 118 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1    10  10      1
2    12  12      1
3    39  43      5
4    55 165     111

```

```
xyz<- pdbfit(pdb,core.inds,outpath="corefit_structures")

rf <- rmsf(xyz)
plotb3(rd,sse=pdb)
```

Warning in plotb3(rd, sse = pdb): Length of input 'sse' does not equal the length of input 'x'; Ignoring 'sse'

```
abline(v=100,col=gray,ylab="RMSF")
```

Warning in int_abline(a = a, b = b, h = h, v = v, unf = unf, ...): supplied color is neither numeric nor character



```
library(jsonlite)
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names=TRUE)
pae1 <- read_json(pae_files[1],simplifyVector=TRUE)
pae5 <- read_json(pae_files[5],simplifyVector=TRUE)
attributes(pae1)
```

```
$names  
[1] "plddt"    "max_pae"  "pae"      "ptm"
```

```
head(pae1$plddt)
```

```
[1] 83.62 88.69 88.25 90.38 90.19 89.88
```

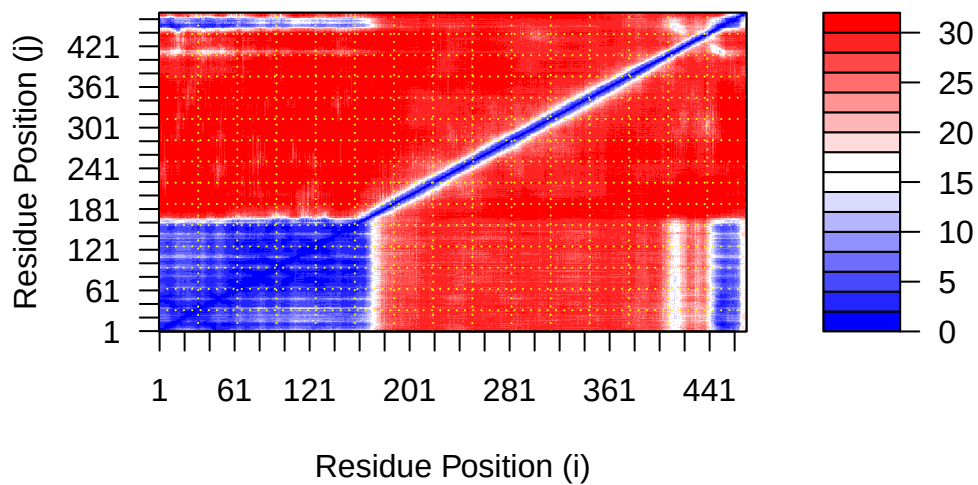
```
pae1$max_pae
```

```
[1] 31.39062
```

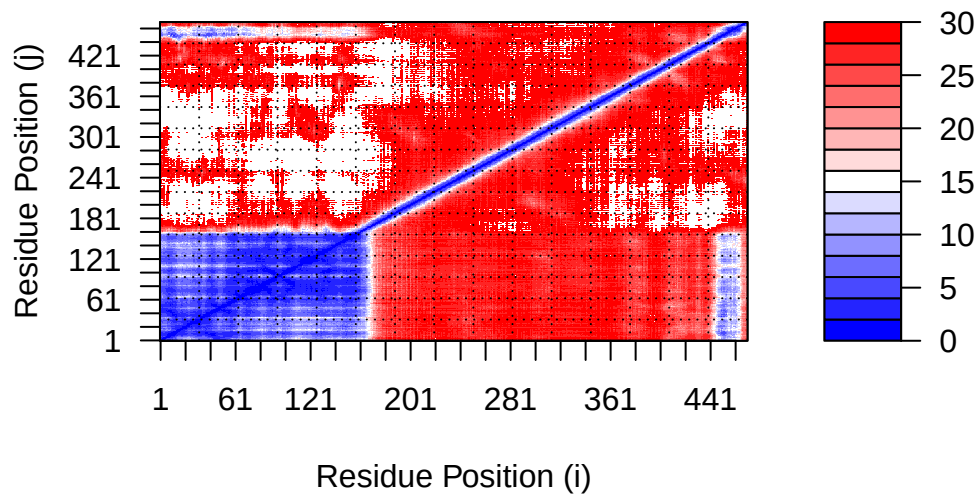
```
pae5$max_pae
```

```
[1] 31.42188
```

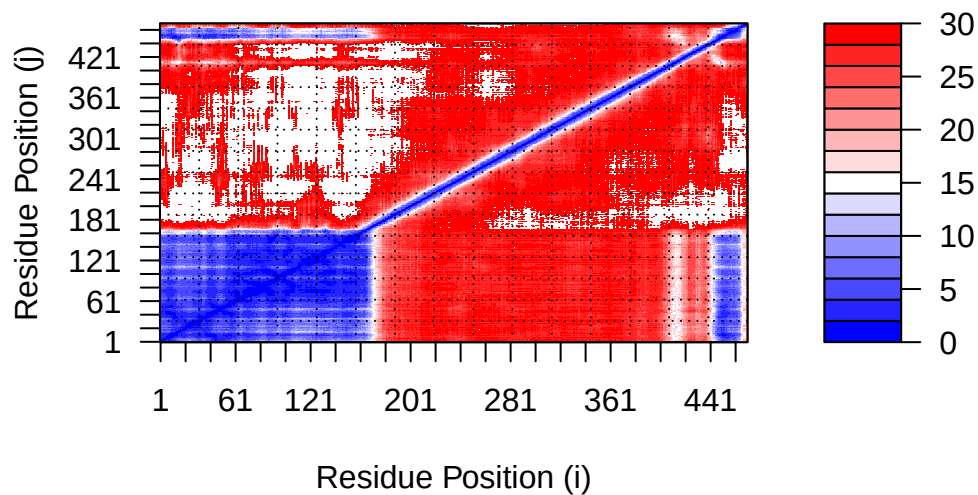
```
plot.dmat(pae1$pae,  
          xlab="Residue Position (i)",  
          ylab="Residue Position (j)")
```




```
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col="black",
          zlim=c(0,30))
```



```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



```
aln_file <- list.files(path=results_dir,
                      pattern=".a3m$",
                      full.names = TRUE)
aln_file
```

```
[1] "novelProtein_7a997/novelProtein_7a997.a3m"
```

```
aln<-read.fasta(aln_file[1],to.upper=TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 21247 591
```

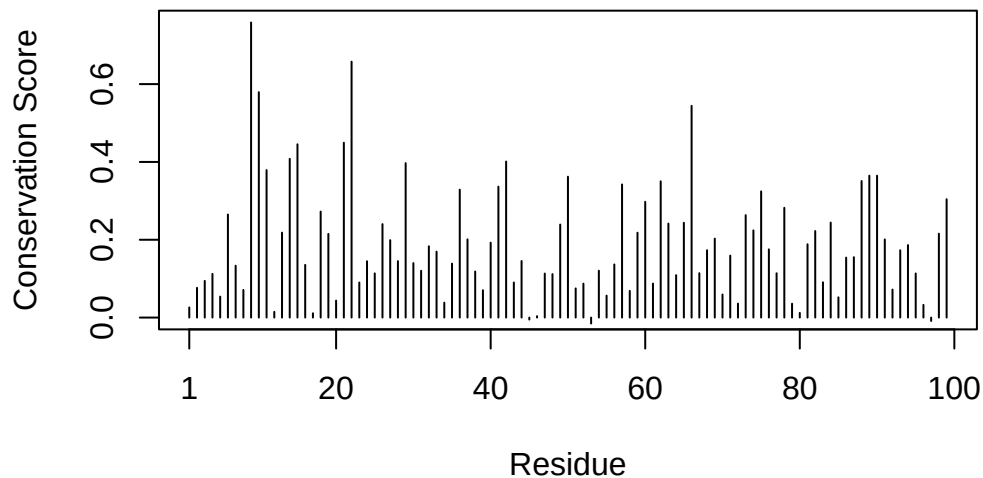
```
sim<-conserv(aln)
```

```
plotb3(sim[1:99],see=trim.pdb(pdb,chain="A"), ylab="Conservation Score")
```

Warning in plot.window(xlim, ylim, ...): "see" is not a graphical parameter

Warning in plot.xy(xy.coords(x, y), type = type, ...): "see" is not a graphical parameter

Warning in title(main = main, sub = sub, xlab = xlab, ylab = ylab, ...): "see" is not a graphical parameter



```
con <- consensus(aln,cutoff=0.9)
con$seq
```

```
[1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[19] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[145] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[163] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[181] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[199] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[217] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
```

```

[235] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[253] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[271] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[289] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[307] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[325] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[343] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[361] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[379] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[397] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[415] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[433] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[451] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[469] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[487] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[505] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[523] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[541] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[559] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[577] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"

```

```

m1.pdb <- read.pdb(pdb_files[1])

# your sim vector
sim_vec <- sim
res_index <- unique(m1.pdb$atom$resno)
#fix length mismatch
sim_res <- rep(sim_vec, length.out = length(res_index))

occ <- sim_res[match(m1.pdb$atom$resno, res_index)]

write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")

```

FIGURE 18

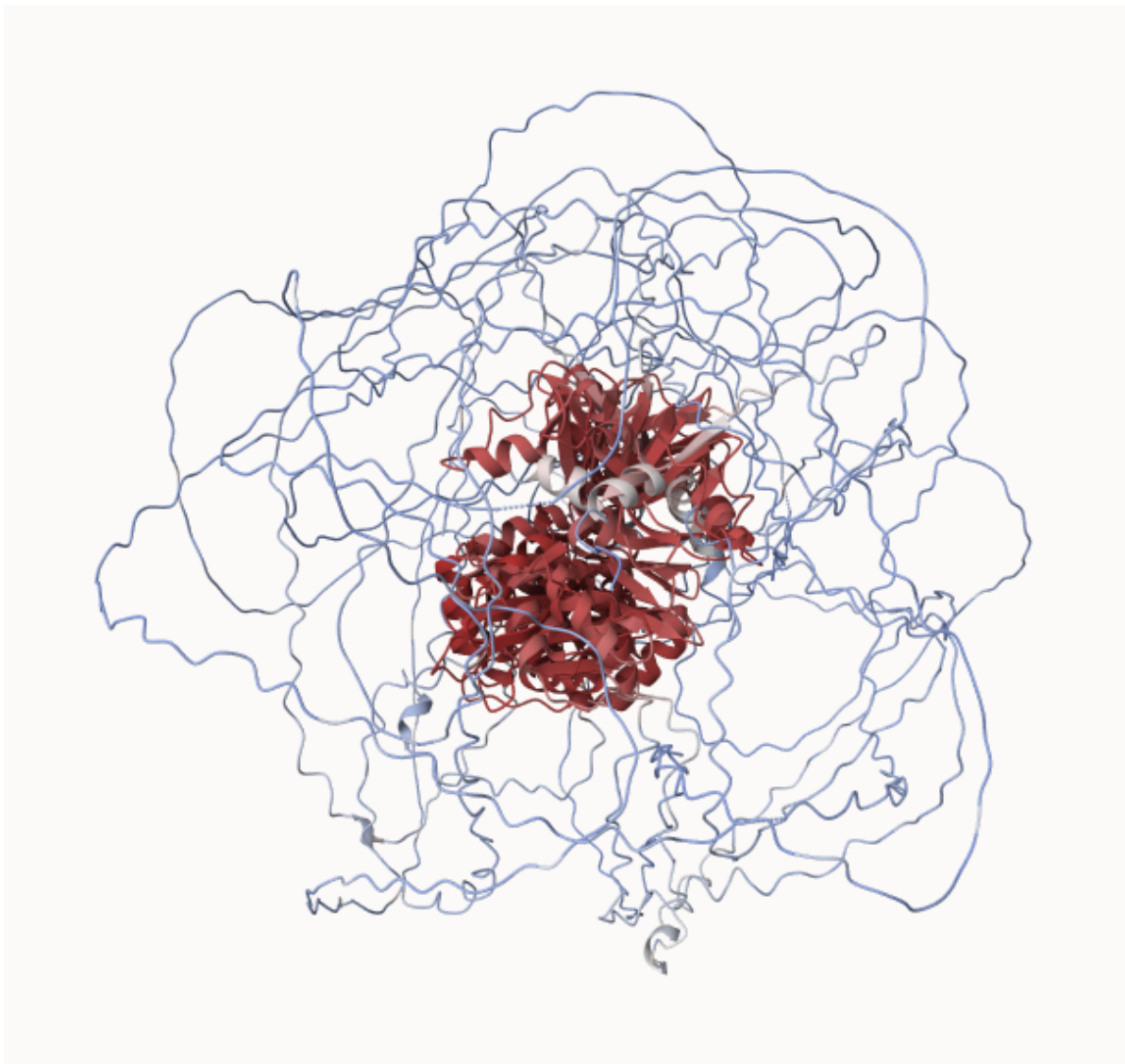


FIGURE 19

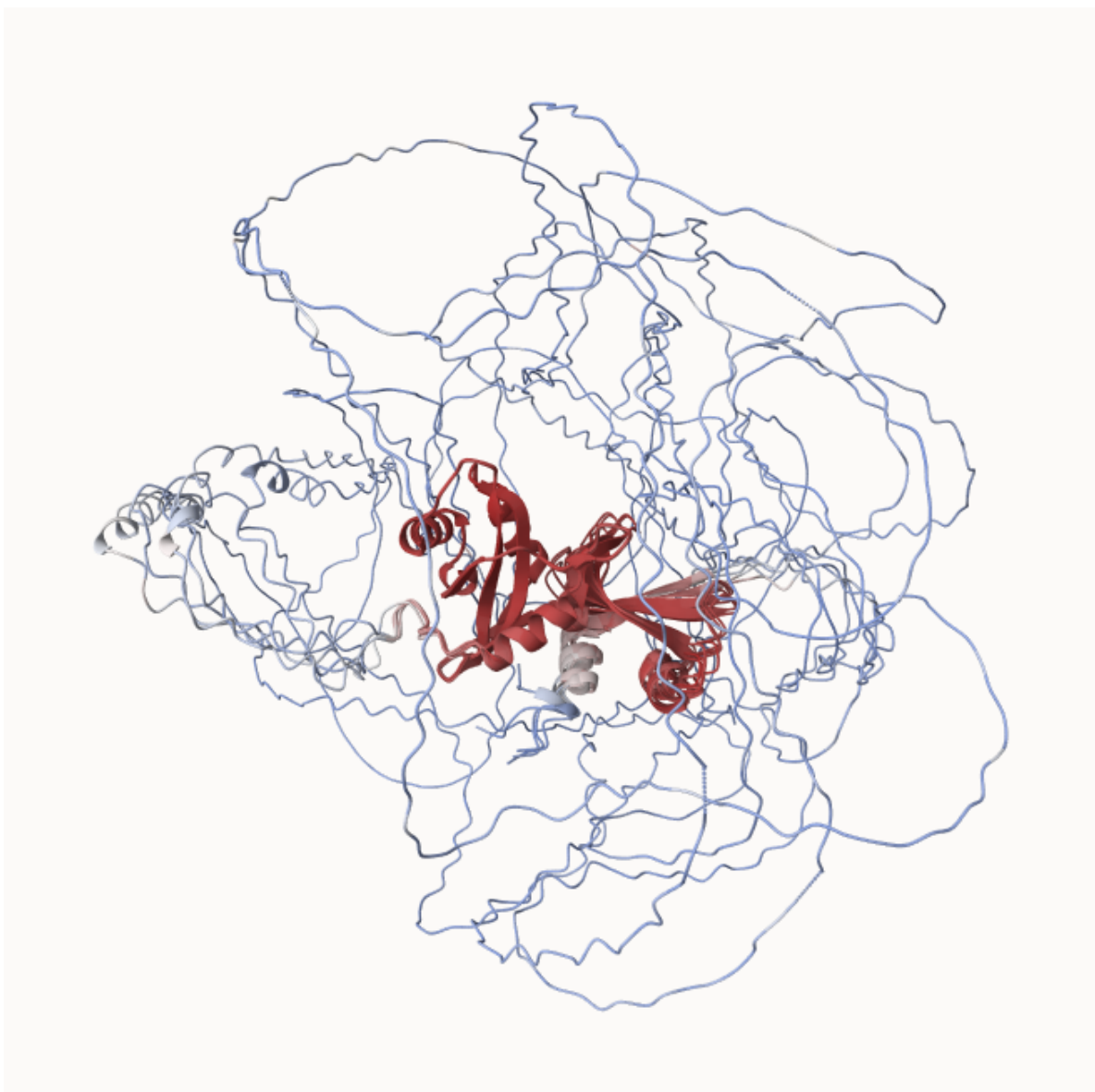


FIGURE 20

